

### General Instructions:

- The submission due date of this assignment is **Thursday (8<sup>th</sup> April 2026) midnight** (before **12:00 am**)
- Write a report (i.e. in a word file) that illustrates your main solution steps including the best fitness values and the average values, plotted over generations (with and without elitism)
- Zip your code and the report in a file entitled [**YourName\_YourID\_AssignmentNumber**]
- This assignment should be delivered and discussed INDIVIDUALLY .

### ➤ Requirements:

**In this assignment it is required to implement a Cuckoo search algorithm to solve the following problem:**

We are minimizing

$$f(x, y) = (x - 2)^2 + (y - 2)^2$$

### Algorithm:

- 1. Initialize nests**
  - Generate random solutions
  - You need:
    - Number of nests:  $n=20$
    - Discovery probability:  $pa=0.25$
    - Step size:  $\alpha=1$
    - Iterations: 100
    - Variables Range =  $[-5: 5]$
- 2. Generate new solutions (via Lévy flights)**
  - Random walk with long jumps
- 3. Evaluate fitness**
  - Check how good each solution is
- 4. Replace worse nests**
  - If a new solution is better → replace
- 5. Abandon some nests**
  - With probability  $pap\_apa$ , discard bad solutions

**6. Keep the best**

Repeat until convergence